

SPIN Codes: Single-Permutation INterleaved Codes

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Abstract

Pseudorandom correlation generators and proof systems need binary codes that combine constant rate and linear distance with fast ordinary and transposed encoding. They also need predictable data movement. Random linear codes provide strong distance but require dense computation. Fast structured encoders reduce this cost, but their sparse dependencies make linear distance difficult to prove. We ask whether one code family can achieve all of these properties.

We introduce Single-Permutation INterleaved (SPIN) codes. A SPIN encoder composes a blockwise outer encoder, one global interleaver, and a recursive inner encoder. Our central result is a Structured SPIN family at native lengths N . Every realized code has rate $1/2$, and its ordinary and transposed encoders use $O(N)$ bit operations. With probability $1 - o(1)$ over setup, its minimum distance exceeds $0.11N$. The construction reuses one sampled spectrum-certified BA-3 constituent along the outer diagonal, factors the interleaver into structured routing stages, and uses the fixed RM2Sub-S19 inner. The outer interface permits different base constituents and base lengths. The current certificate uses extended Golay blocks and the schedule $\frac{39}{4} \log_2 N + O(1)$. Zero extension gives a family at every sufficiently large requested length with rate $1/2 - o(1)$, relative distance greater than $0.11 - o(1)$, and linear work.

The proof uses a finite, class-indexed first-moment framework. The outer stage supplies class multiplicities, while the interleaver-inner pair supplies conditional failure envelopes. Two additional families isolate the main proof mechanisms. At rate $1/2$, Accumulator SPIN achieves relative distance 0.0037 , and Random SPIN achieves relative distance 0.11002 , under explicit logarithmic schedules. A finite-length specialization is under separate constituent selection and certification; it is not used by the scalable theorem.

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1 Introduction

Binary codes inside correlation generators and proof systems are algorithms, not merely stored generator matrices. They must combine constant rate and linear distance with fast ordinary and transposed encoding and regular data movement.

Pseudorandom correlation generators apply a public generator matrix to large vectors to expand sparse correlations into OT and VOLE correlations [BCGI18, BCG⁺19]. Code-based proof systems encode rows of large arrays, with distance supporting soundness [GLS⁺23, BCF⁺25]. Transposition is also operational: for a linear encoder E and check vector q , the identity $\langle E(x), q \rangle = \langle x, E^\top(q) \rangle$ pulls the check back to the message coordinates. At these dimensions, long-range memory traffic can cost more than the binary arithmetic.

Random linear codes provide excellent distance, but their encoders are dense. Fast structured encoders reduce the computation and make memory access more regular. Their sparse dependencies, however, make linear distance difficult to prove. The central question is whether one binary code family can achieve strong distance, linear-time evaluation in both directions, and predictable data movement.

This paper answers that question with Structured SPIN. At its native lengths N , every realized code has rate $1/2$ and $O(N)$ ordinary and transposed encoding work. With probability $1 - o(1)$ over setup, its minimum distance exceeds $0.11N$. One global factored interleaver contains the long-range data movement; the outer permutations remain inside logarithmic-size blocks. A zero-extension wrapper gives rate $1/2 - o(1)$, relative distance $0.11 - o(1)$, and linear work at every sufficiently large requested length.

The construction follows the serial-concatenation pattern. An outer encoder prevents messages from remaining too sparse. A recursive inner encoder spreads the interleaved outer word across the final codeword. The interleaver moves the outer support before the recursion processes it. SPIN makes this pattern modular: structure appears where speed and regular data movement require it, and setup randomness remains at the interfaces used by the distance proof.

For an admissible output length N , a SPIN encoder has the form

$$E_N := I_N \circ \Pi_N \circ O_N. \tag{1}$$

The outer encoder O_N maps the message into N intermediate bits. The interleaver Π_N is one permutation of those N coordinates. The inner encoder I_N is a length-preserving recursive linear map. Setup may sample any of these objects from a declared distribution.

The phrase *single permutation* describes the mathematical interface. An implementation may factor Π_N into block shuffles, a transpose, and region shuffles. These factors still compose to one global interleaver.

1.1 A family of SPIN families

We use the same interface for three construction lines.

Accumulator SPIN. The outer encoder uses random block maps along the diagonal. The interleaver is uniform, and the inner encoder is the binary accumulator. This is the depth-two, one-accumulator truncation of the Block-Accumulate architecture of Kolesnikov et al. [KPR⁺26]. The BA construction also supplies the random-block enumerator used below. Chosen-block BAA instead studies the related option of reusing one certified constituent in every outer position [PR26]. The accumulator admits an exact input–output enumerator, making this construction the simplest setting in which to expose the distance mechanism.

Family	Outer	Interleaver	Inner
Accumulator SPIN	random blocks	uniform	accumulator
Random SPIN	random blocks	uniform	random recursive
Structured SPIN	spectrum-controlled	structured	RM2Sub-S19

Table 1: The three principal SPIN construction lines.

Random SPIN. The outer encoder and uniform interleaver remain unchanged. A random recursive convolution replaces the accumulator. The stronger inner mixing yields a substantially larger relative distance, at the cost of a denser recursion. Silver codes earlier used a banded lower-triangular solve, which is a structured recursive encoder of convolutional form [CRR21]. Expand-Convolute later isolated a random time-varying convolution after a sparse expander [RRT23]. The theorem below combines that random inner with independent random-block injections.

Structured SPIN. A spectrum-controlled local outer, a structured global interleaver, and a structured recursive inner replace the random components. The scalable family samples one spectrum-certified BA-3 constituent and reuses it in every outer position. Its base code and base length are parameters. Independent block and region shuffles surround a deterministic transpose, and RM2Sub-S19 supplies the inner recursion. The direct family has natural admissible lengths. Zero extension from the largest preceding direct member gives a code at every sufficiently large requested length. A separately tuned implementation at $N = 2^{21}$ is a finite specialization, not the family definition.

Table 1 summarizes the three lines.

The component table also defines a larger design space. For example, one may combine a structured interleaver with a random outer and an accumulator. This paper does not analyze every combination. An additional combination enters the main development only when it supplies a necessary proof bridge or a distinct quantitative result.

1.2 Contributions

We make four contributions.

- **A structured linear-time code with proved distance.** We construct a sampled binary code family whose native members have rate $1/2$ and minimum distance greater than $0.11N$ with probability $1 - o(1)$. Every realized member has $O(N)$ ordinary and transposed encoding work. The result therefore combines strong binary distance with linear work in both evaluation directions.
- **A modular structured construction.** Structured SPIN combines one sampled BA-3 constituent reused along the outer diagonal, one factored global interleaver, and the fixed RM2Sub-S19 inner. The base constituent and its length are parameters exposed through a certified spectrum envelope. The interleaver contains the long-range data movement; the remaining outer permutations are block-local. Constituent replacement therefore does not change the route or inner architecture.
- **A first-moment framework for structured routes.** We index outer words by classes and bound each class’s conditional failure probability under the interleaver–inner pair. Equal-

weight words need not have the same inner law. The Structured SPIN proof uses occupation and row-weight classes.

- **Two explanatory SPIN families.** At rate $1/2$, Accumulator SPIN proves relative distance 0.0037 , while Random SPIN proves relative distance 0.11002 . These baselines isolate the block spectrum, uniform interleaving, and recursive mixing.

1.3 Related work

Serially concatenated codes. Turbo codes introduced randomized interleaving as a way to combine simple constituent encoders [BGT93]. Benedetto et al. developed the uniform-interleaver analysis of serial concatenations [BDMP98]. Divsalar, Jin, and McEliece introduced repeat-accumulate codes and their weight-enumerator analysis [DJM98], while Kahale and Urbanke studied minimum distance in random parallel and serial concatenations [KU98]. Bazzi, Mahdian, and Spielman proved limitations for depth-two constant-memory serial concatenations and positive results with two accumulators [BMS09]. Later work established coding theorems for repeat-multiple-accumulate ensembles approaching the binary Gilbert–Varshamov bound [KZKC08].

Block-Accumulate codes replace repetition with random block-diagonal maps and compute detailed expected spectra; they also provide optimized CPU and GPU implementations [KPR⁺26]. Accumulator SPIN is their one-accumulator truncation. Chosen-block BAA studies the complementary choice of reusing one constituent in every diagonal position [PR26]. Structured SPIN adopts that shared-code view, but adds a structured global route and a different recursive inner.

Codes for correlation generation. Linear codes compress sparse correlations in vector-OLE and pseudorandom correlation generators [BCGI18, BCG⁺19]. This application motivated a sequence of faster structured codes. Silver uses structured LDPC matrices and a banded triangular solve [CRR21]. Linear distance was an explicit conjecture supported by heuristic evidence. Expand-Accumulate combines a sparse expander with an accumulator [BCG⁺22]. Expand-Convolute replaces that accumulator with a random time-varying convolution [RRT23]. Random SPIN keeps the latter inner distribution but replaces the expander with random block injections. Structured SPIN instead combines a spectrum-controlled BA outer, one factored global interleaver, and the RM2Sub inner. Its principal distinction is the simultaneous proof of rate one half, relative distance 0.11 , and linear ordinary and transposed encoding work.

Linear-time proving systems. Constant-overhead arguments make linear-time encoding a first-order prover cost [RZR25]. Orion uses code switching to obtain a linear-time prover [XZS22], and Brakedown builds a field-agnostic SNARK from a linear-time encodable code [GLS⁺23]. Field-agnostic SNARKs have also been instantiated from Expand-Accumulate codes [BFK⁺24]. Blaze uses interleaved RAA codes and supplies improved binary distance bounds [BCF⁺25]. SPIN does not instantiate a proof system in this paper. These works motivate the combination of high rate, strong binary distance, fast transposed evaluation, and regular data movement.

1.4 One distance calculation

All three lines use the same first-moment idea. Fix a target distance d , and count nonzero messages whose encoded words have weight at most d . If the expected count is below one, then some sampled encoder is injective and has minimum distance greater than d .

For a uniform interleaver, the outer Hamming weight is often a sufficient interface. The expected bad-word count then has the form

$$\sum_w A_w^{\text{out}} p_w^{\text{in}}(d), \quad (2)$$

where A_w^{out} is the expected outer spectrum and $p_w^{\text{in}}(d)$ is an inner slice-to-tail probability.

A structured interleaver need not produce a uniform Hamming slice. We therefore allow a class to record block occupation or another routing profile. The interleaver–inner analysis supplies a uniform failure envelope for each class. The weight-indexed sum in (2) is the special case in which the conditional law depends only on weight.

1.5 Organization

Section 2 fixes notation. Section 3 states the exact finite first-moment framework. Sections 4 and 5 develop the two random construction lines. Section 6 defines the structured line, its parameterized constituent interface, and the currently certified scalable instantiation. Section 7 specifies finite certificate semantics. Section 8 separates finite validity, asymptotic distance, and encoding complexity. Section 9 records the implementation and finite-evaluation boundary. The appendix records the scalable certificate and finite-length specialization.

2 Preliminaries

2.1 Notation

For an integer $n \geq 0$, let $[n] := \{1, \dots, n\}$, with $[0] := \emptyset$. All vector spaces are over $\mathbb{F}_2$. We write vectors as rows. For $u \in \mathbb{F}_2^n$, let $\text{wt}(u)$ be its Hamming weight and let $\text{supp}(u) := \{i \in [n] : u_i \neq 0\}$ be its support. Let S_n denote the symmetric group on $[n]$; a permutation acts on vectors by permuting their coordinates.

If S is a finite set, then $X \leftarrow S$ denotes a uniform sample from S . If $\mathcal{D}$ is a distribution, then $X \leftarrow \mathcal{D}$ denotes a sample from $\mathcal{D}$. Separate sampling statements use independent randomness unless a joint distribution is declared.

For $q \in [0, 1]$, define the binary entropy function

$$H_2(q) := -q \log_2 q - (1 - q) \log_2 (1 - q),$$

where $0 \log_2 0 := 0$. For integers a, b , we set $\binom{a}{b} := 0$ unless $0 \leq b \leq a$. These conventions shorten the enumerator formulas below.

2.2 Linear encoders and codes

Fix integers $1 \leq K \leq N$. A binary linear encoder is a map

$$E : \mathbb{F}_2^K \longrightarrow \mathbb{F}_2^N.$$

Under the row-vector convention, there is a unique generator matrix $G_E \in \mathbb{F}_2^{K \times N}$ satisfying $E(x) = xG_E$. The associated linear code is the image

$$\mathcal{C}(E) := E(\mathbb{F}_2^K) \subseteq \mathbb{F}_2^N.$$

Its rate is $\dim(\mathcal{C}(E))/N$. In particular, an injective encoder has rate K/N .

For a nonzero linear code $\mathcal{C} \subseteq \mathbb{F}_2^N$, define

$$d_{\min}(\mathcal{C}) := \min_{c \in \mathcal{C} \setminus \{0\}} \text{wt}(c).$$

For bookkeeping in statements that test injectivity separately, set $d_{\min}(\{0\}) := +\infty$, and write $d_{\min}(E) := d_{\min}(\mathcal{C}(E))$. The relative distance is $d_{\min}(\mathcal{C})/N$.

We keep injectivity separate from the distance of the image. A noninjective map can have an image of positive minimum distance, while a nonzero kernel message encodes to zero. Accordingly, a claim that a K -to- N encoder has rate K/N and relative distance δ includes injectivity and the bound $d_{\min}(E) \geq \delta N$.

The transposed encoder is the linear map

$$E^\top : \mathbb{F}_2^N \longrightarrow \mathbb{F}_2^K, \quad E^\top(q) := qG_E^\top.$$

It satisfies $\langle E(x), q \rangle = \langle x, E^\top(q) \rangle$. Ordinary encoding work refers to evaluating E ; transposed work refers to evaluating $E^\top$. Unless stated otherwise, work bounds count bit operations. We discuss long-range data movement separately because the bit-operation count does not capture it.

2.3 Sampled families and asymptotics

A sampled encoder includes its setup distribution. After setup, the realized linear map and its image code are fixed. Expectations and probabilities refer to the declared setup distribution unless a subscript specifies a smaller probability space.

For a family indexed by native lengths $N_m \rightarrow \infty$, the notation $o(1)$ refers to the limit $m \rightarrow \infty$. An event holds *with high probability* if its failure probability is $o(1)$ over setup. A deterministic wrapper may map these native members to a denser set of requested lengths.

3 A common first-moment framework

This section isolates the calculation shared by all SPIN families. We first define the complete sampled encoder. We then count low-weight outputs of nonzero messages.

Fix lengths $K, N \geq 1$. The construction samples an outer encoder $O : \mathbb{F}_2^K \rightarrow \mathbb{F}_2^N$ and a pair (Π, I) , where Π is a permutation of $[N]$ and $I : \mathbb{F}_2^N \rightarrow \mathbb{F}_2^N$ is linear. Assume that O is independent of (Π, I) . The realized SPIN encoder is

$$E := I \circ \Pi \circ O.$$

Fix a finite class set $\mathcal{T}$ and a class map $c : \mathbb{F}_2^N \rightarrow \mathcal{T}$. The class map records the information about an outer word that remains relevant after the outer stage.

For a fixed outer word $u \in \mathbb{F}_2^K$, define its conditional failure probability

$$q_d(u) := \mathbb{P}_{\Pi, I}[\text{wt}(I(\Pi(u))) \leq d]. \quad (3)$$

The probability is over the interleaver and inner setup; the word u is fixed. For every class $\tau \in \mathcal{T}$, let $Q_\tau(d)$ be any certified envelope satisfying

$$q_d(u) \leq Q_\tau(d) \quad \text{for every } u \text{ with } c(u) = \tau. \quad (4)$$

Set $Q_\tau(d) := 0$ when the class has no preimage. The class need not determine the exact failure probability. It need only expose enough information to prove the uniform bound in (4).

For $\tau \in \mathcal{T}$, define the expected outer multiplicity

$$A_\tau^{\text{out}} := \mathbb{E}_O \left[\sum_{x \in \mathbb{F}_2^K \setminus \{0\}} \mathbf{1}\{c(O(x)) = \tau\} \right]. \quad (5)$$

Theorem 3.1 (Class-indexed first-moment envelope). *For $d \in \{0, \dots, N\}$, define*

$$Z_d := |\{x \in \mathbb{F}_2^K \setminus \{0\} : \text{wt}(E(x)) \leq d\}|.$$

Then

$$\begin{aligned} \mathbb{P}[E \text{ is not injective or } d_{\min}(E(\mathbb{F}_2^K)) \leq d] &\leq \mathbb{E}[Z_d] = \mathbb{E}_O \left[\sum_{x \in \mathbb{F}_2^K \setminus \{0\}} q_d(O(x)) \right] \\ &\leq \sum_{\tau \in \mathcal{T}} A_\tau^{\text{out}} Q_\tau(d). \end{aligned} \quad (6)$$

If q_d is constant on every class and $Q_\tau(d)$ equals that constant, then the final inequality is an equality.

Proof. The bad event in (6) occurs exactly when $Z_d \geq 1$. Indeed, a nonzero kernel message contributes output weight zero, and every nonzero image word of weight at most d has a nonzero preimage. Markov's inequality gives $\mathbb{P}[Z_d \geq 1] \leq \mathbb{E}[Z_d]$.

For the identity, expand Z_d as a sum over nonzero messages and condition on the outer realization. Independence of O from (Π, I) gives

$$\mathbb{E}[Z_d] = \mathbb{E}_O \left[\sum_{x \neq 0} q_d(O(x)) \right].$$

Group the terms by $\tau = c(O(x))$ and apply (4). Equation (5) gives the final inequality. Equality holds under the condition in the theorem statement. $\square$

If $\mathcal{G}$ is an event determined only by the outer setup, the theorem also applies under the conditional law $O \mid \mathcal{G}$. The pair (Π, I) remains independent of the conditioned outer encoder. This observation lets a proof first select an outer spectrum event and then certify route envelopes on that event.

3.1 Uniform-interleaver specialization

Suppose Π is uniform over all permutations of $[N]$. If u has weight w , then $\Pi(u)$ is uniform over the Hamming slice

$$\mathcal{H}_{N,w} := \{v \in \mathbb{F}_2^N : \text{wt}(v) = w\}.$$

Indeed, every vector in $\mathcal{H}_{N,w}$ has exactly $w!(N-w)!$ preimages among the permutations of u .

Let U_w be uniform over $\mathcal{H}_{N,w}$, independent of the inner setup, and define

$$p_w^{\text{in}}(d) := \mathbb{P}_{U_w, I}[\text{wt}(I(U_w)) \leq d]. \quad (7)$$

Also define

$$A_w^{\text{out}} := \mathbb{E}_O[|\{x \neq 0 : \text{wt}(O(x)) = w\}|].$$

Corollary 3.2 (Weight-indexed first moment). *Under a uniform interleaver independent of O and I ,*

$$\mathbb{E}[Z_d] = \sum_{w=0}^N A_w^{\text{out}} p_w^{\text{in}}(d).$$

Accumulator SPIN and Random SPIN use Corollary 3.2. Structured SPIN uses Theorem 3.1 with occupation and row-weight classes. The structured-route analysis supplies a uniform failure envelope for every such class; it does not require the conditional probability to be identical for all words in the class.

4 Accumulator SPIN

Accumulator SPIN is the transparent instantiation of the framework. It uses a random block-diagonal outer, a uniform interleaver, and the binary accumulator.

4.1 Relation to Block-Accumulate codes

Block-Accumulate (BA) codes concatenate a random block-diagonal outer with one or more stages, each consisting of a uniform interleaver and an accumulator [KPR⁺26]. Accumulator SPIN retains the first such stage and therefore may be viewed as a truncated BA code. We retain the SPIN name to place this construction in the common outer–interleaver–inner interface, but the block-accumulate architecture and the random-block enumerator below are due to Kolesnikov et al.

Accumulator SPIN follows the BA ensemble at this level: setup samples the local maps independently. The theorem below conditions each local map to be injective. This condition makes every realized outer encoder injective.

Chosen-block BAA studies the related fixed-constituent choice [PR26]. It fixes a short constituent code and uses the same code in every outer position. Its concrete binary constituents include Golay, Reed–Muller, and a BCH subcode. That reuse option is not the ensemble analyzed here.

4.2 Random-block outer lineage

The following local input–output enumerator is the random-block calculation used for BA codes. We state it in the present notation because it is also the local input to the Accumulator SPIN analysis.

Lemma 4.1 (Uniform random-block enumerator). *Suppose that G_B is uniform over $\mathbb{F}_2^{K_B \times B}$. Let $E_{a,b}^{\text{blk}}(G_B)$ be the number of inputs of weight a whose images have weight b . Then*

$$\mathbb{E}_{G_B}[E_{a,b}^{\text{blk}}(G_B)] = \begin{cases} 1, & a = 0 \text{ and } b = 0, \\ 0, & a = 0 \text{ and } b > 0, \\ \binom{K_B}{a} \binom{B}{b} 2^{-B}, & a > 0. \end{cases} \quad (8)$$

Proof. The zero input maps to the zero output. Fix a nonzero input u . Each column of G_B is uniform and independent, so the coordinates of uG_B are independent uniform bits. Thus uG_B is uniform over $\mathbb{F}_2^B$. There are $\binom{K_B}{a}$ inputs of weight a and $\binom{B}{b}$ outputs of weight b . Summing the probability 2^{-B} over these pairs gives (8). $\square$

The BA independent-block ensemble has a closed global input–output enumerator. Let $E_{a,b}^{\text{BA}}$ count total input weight a and total outer-output weight b across M blocks. Specializing the BA formula [KPR⁺26] to local dimensions $K_B \times B$ gives

$$\mathbb{E}[E_{a,b}^{\text{BA}}] = \sum_{q=0}^M 2^{-Bq} \binom{M}{q} \binom{Bq}{b} \sum_{j=0}^q (-1)^j \binom{q}{j} \binom{K_B(q-j)}{a}. \quad (9)$$

Here q is the number of nonzero input blocks. The inner alternating sum counts weight- a inputs for which exactly those q blocks are nonzero. For a fixed such input, the q independently sampled local matrices produce a uniform word on the corresponding Bq output coordinates. This yields the factor $2^{-Bq} \binom{Bq}{b}$.

The theorem below uses the same independence pattern but samples uniform injections rather than unconditioned matrices. Its exact generating function appears in Section 4.4.

4.3 Accumulator inner

Define $\text{Acc}_N : \mathbb{F}_2^N \rightarrow \mathbb{F}_2^N$ by

$$(\text{Acc}_N(u))_t := \sum_{i=1}^t u_i \pmod{2}, \quad t \in [N]. \quad (10)$$

Equivalently, if $y := \text{Acc}_N(u)$ and $y_0 := 0$, then $y_t = y_{t-1} \oplus u_t$.

For $w, h \in \{0, \dots, N\}$, define

$$A_{w,h}^{\text{Acc}} := |\{u \in \mathbb{F}_2^N : \text{wt}(u) = w, \text{wt}(\text{Acc}_N(u)) = h\}|.$$

Lemma 4.2 (Exact accumulator enumerator). *Fix $w \in \{1, \dots, N-1\}$ and set $k := \lceil w/2 \rceil$. Then*

$$A_{w,h}^{\text{Acc}} = \binom{h-1}{k-1} \binom{N-h}{w-k}. \quad (11)$$

Proof. Let the ones of u occur at $1 \leq p_1 < \dots < p_w \leq N$. Set $p_0 := 0$, $p_{w+1} := N+1$, and $\Delta_i := p_i - p_{i-1}$. The positive spacings satisfy $\sum_{i=1}^{w+1} \Delta_i = N+1$.

The accumulator output equals one between its first and second flips, between its third and fourth flips, and so on. Hence its weight is the sum of the even-index spacings. There are k such spacings. The number of positive compositions of h into those k parts is $\binom{h-1}{k-1}$. The remaining $w+1-k$ spacings sum to $N+1-h$, which gives $\binom{N-h}{w-k}$ choices. Multiplication proves (11). $\square$

If U_w is uniform over the weight- w slice, then

$$\mathbb{P}[\text{wt}(\text{Acc}_N(U_w)) \leq d] = \frac{1}{\binom{N}{w}} \sum_{h=0}^d A_{w,h}^{\text{Acc}}. \quad (12)$$

Lemma 4.3 (A simple accumulator tail). *Fix $\delta \in (0, 1/4)$ and $w \in \{1, \dots, \lfloor N/2 \rfloor\}$. Then*

$$\mathbb{P}[\text{wt}(\text{Acc}_N(U_w)) \leq \lfloor \delta N \rfloor] \leq (4e\delta)^{\lceil w/2 \rceil}.$$

Proof. Set $k := \lceil w/2 \rceil$. Lemma 4.2 and the hockey-stick identity give

$$\sum_{h \leq \lfloor \delta N \rfloor} A_{w,h}^{\text{Acc}} \leq \binom{N}{w-k} \binom{\lfloor \delta N \rfloor}{k}.$$

After division by $\binom{N}{w}$, bound the binomial term by $(e\delta N/k)^k$. Since $w \leq N/2$, the remaining binomial ratio is at most $(2w/N)^k$. Finally, $w \leq 2k$, which gives the claim. $\square$

The next contraction is uniform in the input weight. It is the form used by the asymptotic theorem.

Lemma 4.4 (Sharp accumulator contraction). *Fix $\delta \in (0, 1/2)$ and define*

$$\rho(\delta) := 2\sqrt{\delta(1-\delta)}.$$

For every $w \in \{1, \dots, N\}$,

$$\mathbb{P}[\text{wt}(\text{Acc}_N(U_w)) \leq \lfloor \delta N \rfloor] \leq \rho(\delta)^w. \quad (13)$$

Proof. Set $D := \lfloor \delta N \rfloor$ and $k := \lceil w/2 \rceil$. For $w < N$, Lemma 4.2 identifies the probability in (13) with $\mathbb{P}[X \geq k]$, where X is hypergeometric with sample size w and D marked positions among N . For $t > 0$, the moment bound for sampling without replacement gives

$$\mathbb{E}[e^{tX}] \leq (1 - \delta + \delta e^t)^w.$$

Markov's inequality and $k \geq w/2$ give

$$\mathbb{P}[X \geq k] \leq \inf_{t>0} \left(e^{-t/2} (1 - \delta + \delta e^t) \right)^w.$$

The minimizer satisfies $e^t = (1 - \delta)/\delta$, which yields the claim. When $w = N$, the accumulator output has weight $\lceil N/2 \rceil > D$, so the left side is zero. $\square$

4.4 Asymptotic theorem

Fix a rational rate $R \in (0, 1)$ and let

$$\mathcal{B}_R := \{B \geq 1 : RB \in \mathbb{Z}\}.$$

For $B \in \mathcal{B}_R$ and each $i \in [M]$, setup samples an independent uniform linear injection

$$C_i : \mathbb{F}_2^{RB} \longrightarrow \mathbb{F}_2^B.$$

It then samples $\Pi \leftarrow S_N$, where $N := MB$. For $x = (x^{(1)}, \dots, x^{(M)})$, define

$$E_N^{\text{acc}}(x) := \text{Acc}_N \left(\Pi(C_1(x^{(1)})) \parallel \dots \parallel C_M(x^{(M)}) \right). \quad (14)$$

The setup remains fixed for all messages. Every realized encoder is injective.

For one local injection, define

$$G_{B,R}(z) := (2^{RB} - 1) \frac{(1+z)^B - 1}{2^B - 1}. \quad (15)$$

Every nonzero input has a uniform nonzero image, so independence across block positions gives

$$\mathbb{E} \left[1 + \sum_{w=1}^N A_{N,w}^{\text{out}} z^w \right] = (1 + G_{B,R}(z))^M. \quad (16)$$

Define

$$\lambda_A(R, \delta) := 1 - R - \log_2(1 + \rho(\delta)). \quad (17)$$

Fix $c > 0$. For each sufficiently large M , let B_M be the least member of $\mathcal{B}_R$ satisfying

$$B_M \geq c \log_2(MB_M),$$

and set $N_M := MB_M$.

Theorem 4.5 (Accumulator SPIN). *Suppose*

$$\lambda_A(R, \delta) > 0, \quad c\lambda_A(R, \delta) \geq 1.$$

Then

$$\mathbb{P}[d_{\min}(E_{N_M}^{\text{acc}}) \leq \delta N_M] = o(1) \quad (M \rightarrow \infty). \quad (18)$$

The family has rate R , block size $\Theta(\log N_M)$, and $\Theta(N_M \log N_M)$ direct encoding cost.

Proof. Let $D := \lfloor \delta N_M \rfloor$ and let Z_D count nonzero messages whose encoded words have weight at most D . Corollary 3.2, Lemma 4.4, and (16) give

$$\mathbb{E}[Z_D] \leq (1 + G_{B_M, R}(\rho(\delta)))^M - 1.$$

For $z > 0$, equation (15) gives

$$G_{B, R}(z) \leq 2 \left(2^{R-1} (1 + z) \right)^B.$$

Set $\lambda := \lambda_A(R, \delta)$. The block schedule implies

$$MG_{B_M, R}(\rho(\delta)) \leq \frac{2}{B_M} N_M^{1-c\lambda} = o(1).$$

Thus $\mathbb{E}[Z_D] = o(1)$, and Markov's inequality proves (18). $\square$

At rate $R = 1/2$, the choice $c = 3$ and $\delta = 37/10000$ satisfies the theorem with a strict interval-certified margin. The corresponding relative-distance guarantee is small but positive. For this proof, the largest possible rate-half limit as $c \rightarrow \infty$ is $0.0449101394 \dots$

5 Random SPIN

Random SPIN replaces the accumulator with a random recursive convolution. Its outer uses the independent random-block injections from Section 4.

5.1 Relation to Silver and Expand-Convolute

Silver codes predate Expand-Convolute and use a banded lower-triangular solve for linear-time encoding [CRR21]. The solve is an earlier structured recursive encoder of convolutional form. Silver does not sample independent convolution taps at every time step.

Expand-Convolute makes the random convolution explicit. It combines a sparse expander with a non-wrapping recursion whose time-dependent feedback coefficients are independent uniform bits [RRT23]. The Random SPIN theorem retains that inner distribution and replaces the expander with independent random-block injections followed by one uniform interleaver.

5.2 Random recursive inner

Fix a memory $m \geq 1$. For every $t \in [N]$, setup samples

$$\alpha_t \leftarrow \mathbb{F}_2^m$$

independently. Set $y_j := 0$ for $j \leq 0$. On input $u \in \mathbb{F}_2^N$, define

$$y_t := u_t + \langle \alpha_t, (y_{t-1}, \dots, y_{t-m}) \rangle, \quad t \in [N]. \quad (19)$$

Write $I_{N, m}^{\text{rnd}}(u) := y$. Once setup fixes $(\alpha_t)_{t=1}^N$, this map is linear and invertible because y_t contains u_t with coefficient one.

5.3 Exact convolution transfer

The random inner admits an exact finite-state transform. Let X mark input weight and let S mark output weight. Index the matrix rows and columns by $z \in \{0, \dots, m\}$, where z is the number of trailing output zeros, clipped at m . Thus $z = m$ means that the convolution state is zero.

Define the nonnegative matrix $T_m(X, S)$ by

$$T_m(X, S)[z, z'] := \begin{cases} (1+X)S/2, & z < m, z' = 0, \\ (1+X)/2, & z < m, z' = z+1, \\ XS, & z = m, z' = 0, \\ 1, & z = m, z' = m, \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

Let e_m be the unit vector at index m , and let $\mathbf{1}$ be the all-ones vector.

Lemma 5.1 (Exact random-convolution transform). *For the setup in (19),*

$$\mathbb{E} \left[\sum_{u \in \mathbb{F}_2^N} X^{\text{wt}(u)} S^{\text{wt}(I_{N,m}^{\text{rnd}}(u))} \right] = e_m^\top T_m(X, S)^N \mathbf{1}. \quad (21)$$

Proof. Suppose $z < m$. The convolution state is nonzero, so the fresh uniform α_t makes the output bit uniform for either input bit. Summing the two input choices with weights 1 and X gives the first two nonzero transitions in (20). If $z = m$, input zero stays in the zero state, while input one produces output one and moves to state zero. These are the remaining two transitions. Multiplying the one-step transforms and summing over the terminal state proves the identity. $\square$

Let U_h be uniform on the weight- h slice, independently of the inner setup. For $0 < S < 1$, $a > 0$, and $D \in \{0, \dots, N\}$, Lemma 5.1 gives

$$\mathbb{P}[\text{wt}(I_{N,m}^{\text{rnd}}(U_h)) \leq D] \leq S^{-D} a^{-h} \frac{e_m^\top T_m(a, S)^N \mathbf{1}}{\binom{N}{h}}. \quad (22)$$

The first step is the lower-tail moment bound. The second bounds the X^h coefficient by evaluating the nonnegative polynomial at a .

For $\delta \in (0, 1/2)$, define

$$p_{N,h}(\delta) := \mathbb{P}[\text{wt}(I_{N,m}^{\text{rnd}}(U_h)) \leq \lfloor \delta N \rfloor]. \quad (23)$$

5.4 Asymptotic theorem

For an even block length B dividing N , set $M := N/B$. Independently sample uniform linear injections

$$C_i : \mathbb{F}_2^{B/2} \longrightarrow \mathbb{F}_2^B, \quad i \in [M],$$

one permutation $\Pi \leftarrow S_N$, and the convolution setup above. Define

$$E_{N,m}^{\text{rnd}}(x) := I_{N,m}^{\text{rnd}} \left(\Pi(C_1(x^{(1)}) \parallel \dots \parallel C_M(x^{(M)})) \right). \quad (24)$$

All setup objects are independent and remain fixed for all messages.

For each sufficiently large M , let B_M^* be the least positive even integer satisfying

$$B_M^* \geq 17 \log_2(MB_M^*).$$

Set

$$N_M^* := MB_M^*, \quad m_M^* := \left\lceil \frac{51}{50} \log_2 N_M^* \right\rceil, \quad \delta_* := \frac{5501}{50000}.$$

Theorem 5.2 (Random SPIN). *For the ensemble in (24),*

$$\mathbb{P}[d_{\min}(E_{N_M^*, m_M^*}^{\text{rnd}}) \leq \delta_* N_M^*] = o(1) \quad (M \rightarrow \infty). \quad (25)$$

The family has rate $1/2$, relative distance greater than 0.11002 , and $\Theta(N_M^ \log N_M^*)$ direct encoding cost.*

Proof. Let $A_{N,h}^{\text{out}}$ be the number of outer words of weight h , and let $p_{N,h}(\delta_*)$ be the probability in (22) with $D = \lfloor \delta_* N \rfloor$. Split the first-moment sum at $h = N/5000$ and $h = 0.9N$.

Lemma B.3 shows that the first range is $o(1)$. Lemma B.2 is uniform for $h/N \in [1/5000, 0.9]$, so the middle range is $2^{-\Omega(N)}$. For the final range, use $p_{N,h} \leq 1$ and the rate-half random-block outer exponent:

$$\sum_{h \geq 0.9N} \mathbb{E}[A_{N,h}^{\text{out}}] \leq 2^{N(H_2(0.9) - 1/2 + o(1))} = 2^{-\Omega(N)}.$$

The complete first moment tends to zero. Markov's inequality proves (25). $\square$

The admissible lengths have relative gaps $o(1)$. For a requested length, zero extend the largest preceding native member. This preserves injectivity and absolute minimum distance while changing rate and relative distance by only $o(1)$.

6 Structured SPIN

6.1 From random mixing to structured mixing

Random SPIN is a useful starting point. Its random outer blocks have a simple expected spectrum. Its uniform interleaver erases coordinate information, and its random recursive inner repeatedly introduces fresh mixing. These properties make the first-moment proof relatively clean.

They also make the encoder expensive. The random recursive inner performs fresh random mixing at every coordinate. Under the parameter schedule used in Section 5, its direct implementation costs more than linear time. Fast encoders usually obtain their speed from repeated circuits, fixed local maps, and predictable data movement. In this sense, fast codes are usually highly structured.

Structured SPIN tries to retain the randomness required by the distance proof while placing structure wherever the encoder needs speed. A useful way to understand this balance is to follow a sparse message through the construction. Sparse messages are not always the only difficult regime: medium-weight messages can also matter because there are many of them. Nevertheless, a message supported in one or two outer blocks is a useful model of the construction's most fragile behavior. Such a message begins with little support, so every later stage must prevent that support from remaining localized or disappearing.

Suppose first that the message activates one outer block. Random SPIN would encode this block with a random injection. To obtain a faster and more explicit construction, we replace that injection with a constituent code whose low-weight spectrum can be certified. The minimum-weight codewords are important, but the proof must also control the nearby low-weight shells, which may

contain many more words. The required outer interface is therefore a certified envelope for the relevant low-weight spectrum, not merely a minimum-distance bound.

Weight alone does not yet describe a structured block output. Two words of equal weight may occupy different coordinates, and those coordinate patterns may behave differently in later stages. Structured SPIN therefore samples an independent output permutation for every outer block. Conditioned on the output weight, the permuted support is uniform within that block. Independence across blocks also prevents a message from deliberately aligning a repeated coordinate pattern across several active blocks. After this local randomization, the first-moment argument can again describe a block output by its weight.

The next question is where this support enters the recursive inner. A uniform global interleaver can, with small probability, send most of an active block near the end of the inner computation. The state then has too few remaining steps in which to spread the support. One can reduce this risk by forcing each outer block to have large output weight, but larger or denser outer blocks increase the encoding cost.

Structured SPIN instead builds the desired spreading property into the interleaver. Its global permutation is factored into simple stages. The within-block permutation first removes the constituent code’s coordinate structure. A deterministic transpose then distributes the coordinates of each block among regions spanning the inner computation. Finally, independent shuffles within those regions randomize the precise locations. The transpose is the structured part that rules out a deliberately late start in the declared routing pattern. The region shuffles stop the message from controlling the remaining alignment and supply the probability law needed by the first-moment calculation.

At this point, we have replaced the random outer and the uniform global interleaver. The remaining expensive component is the random recursive inner. A direct random inner refreshes its mixing at every coordinate. Structured SPIN instead amortizes that refresh over a group of coordinates. During the cheap part of an inner step, fixed structured maps mix one input group with a small persistent state and emit a group of output bits. These maps are fast because they operate on whole groups and reuse a compact circuit. At the refresh stage, multiplication by a sampled field element supplies the uniform-state law used by the proof for each fixed nonzero state. The analysis combines this refresh with certified information about the kernels and images of the fixed maps.

The resulting encoder is neither fully random nor fully deterministic. Its constituent codes and batched inner operations provide speed. Its independent block permutations, region shuffles, and state refreshes supply randomness at the interfaces where the distance proof uses it. The rest of this section turns this design story into a parameterized construction.

6.2 Construction at a glance

We first define one direct member of the family. Choose a constituent length b , a constituent dimension $k \leq b$, a number L of outer-block positions, an inner step width t , and a state dimension s , with $t \mid L$. Set

$$N := Lb, \quad K := Lk, \quad e := L/t, \quad R := N/t = be. \quad (26)$$

Thus the interleaver will have b regions, each region will contain L coordinates, and the inner will process each region in e steps.

The encoder has the following dataflow:

$$\mathbb{F}_2^{Lk} \xrightarrow{O} (\mathbb{F}_2^b)^L \xrightarrow{\Pi_{\text{blk}}} (\mathbb{F}_2^b)^L \xrightarrow{T} (\mathbb{F}_2^L)^b \xrightarrow{\Pi_{\text{reg}}} (\mathbb{F}_2^L)^b \xrightarrow{\text{serialize}} \mathbb{F}_2^N \xrightarrow{I} \mathbb{F}_2^N. \quad (27)$$

The next three subsections define these arrows. The block-diagonal outer O controls the possible weights produced by each active message block. The within-block permutations Π_{blk} erase the coordinate patterns of those outputs. The transpose T spreads every outer block across the b regions, and the region permutations Π_{reg} randomize the locations within each region. The recursive inner I then processes the serialized regions in t -bit steps. In short, the outer supplies a certified weight profile, the route supplies a controlled support law, and the inner turns that routed support into output weight.

There are two kinds of construction data. The fixed data are the integers (L, b, k, t, s) , two linear maps

$$A : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^t, \quad C : \mathbb{F}_2^t \rightarrow \mathbb{F}_2^s, \quad C \circ A = 0,$$

a basis identifying $\mathbb{F}_2^s$ with $\mathbb{F}_2^s$, and a joint setup law for injective constituent maps

$$G_j : \mathbb{F}_2^k \rightarrow \mathbb{F}_2^b \quad (j \in [L]).$$

This joint law must state whether the positions use one fixed constituent, one shared sampled constituent, or independently sampled constituents. The distinction changes the expected outer enumerator.

Setup samples the constituent maps from this joint law. Independently, it samples one uniform coordinate permutation for every outer block, one uniform permutation for every region, and one uniform nonzero field multiplier for every inner step. All routing permutations and inner multipliers are mutually independent. After setup, every arrow in (27) is fixed. A family selector θ_m is a sequence of these parameter choices and setup laws; we suppress m until the scalable instantiation in Section 6.7.

6.3 Block-diagonal outer

Write a message as $u = (u_1, \dots, u_L)$ with $u_j \in \mathbb{F}_2^k$. The outer encoder is the direct sum

$$O(u) := (G_1(u_1), \dots, G_L(u_L)) \in (\mathbb{F}_2^b)^L. \quad (28)$$

Equivalently, a generator matrix for O places the local generator matrices on the diagonal. The most basic instantiation repeats one constituent $G : \mathbb{F}_2^k \rightarrow \mathbb{F}_2^b$ in every position. The scalable construction below builds this shared map from a parameterized base constituent and two accumulator layers. The current certificate uses an extended Golay base constituent, but the construction interface does not fix that choice or its length.

The outer proof interface is its local weight enumerator. For a realized local map G_j , define

$$P_{G_j}(z) := \sum_{v \in \mathbb{F}_2^k \setminus \{0\}} z^{\text{wt}(G_j(v))}. \quad (29)$$

If one deterministic map G is repeated, the number of messages with exactly q active blocks and outer weight h is

$$\binom{L}{q} [z^h] P_G(z)^q. \quad (30)$$

For independently sampled, identically distributed local maps, expectation replaces P_G in (30) by $\mathbb{E}[P_{G_1}]$. If one sampled map is reused in all positions, the correct quantity is instead

$$\binom{L}{q} \mathbb{E}[[z^h] P_G(z)^q]. \quad (31)$$

In particular, an expectation over one sampled constituent must not be moved inside the q th power.

The distance proof need not know how the local encoder is implemented. It may consume an exact enumerator, an expected enumerator under the declared setup, or a coefficientwise certified envelope. Minimum distance alone is generally insufficient because nearby low-weight shells can have much larger multiplicity.

6.4 Structured interleaver

Index the outer coordinates by pairs $(j, r) \in [L] \times [b]$, where j is the outer-block position. For every $j \in [L]$, setup samples $\sigma_j \leftarrow S_b$ and defines

$$\Pi_{\text{blk}}(j, r) := (j, \sigma_j(r)).$$

The deterministic transpose changes the coordinate view by

$$T(j, r) := (r, j) \in [b] \times [L].$$

For every region $r \in [b]$, setup samples $\pi_r \leftarrow S_L$ and defines

$$\Pi_{\text{reg}}(r, j) := (r, \pi_r(j)).$$

The complete interleaver is

$$\Pi := \Pi_{\text{reg}} \circ T \circ \Pi_{\text{blk}}. \quad (32)$$

We serialize the output of (32) in region-major order. Thus region r occupies the consecutive coordinates $(r-1)L+1, \dots, rL$.

Each outer block contributes exactly one coordinate to each of the b regions. Conditional on the local output weight, σ_j makes its active coordinates a uniform subset of the regions. Conditional on every region weight, π_r makes the support uniform within that region. These two symmetries remove the original coordinate pattern while preserving the deterministic spreading supplied by the transpose.

6.5 Structured recursive inner

The inner processes the serialized interleaver output in t -bit steps. Write that output as

$$X = (X_0, \dots, X_{R-1}), \quad X_i \in \mathbb{F}_2^t, \quad R := N/t = be.$$

The state space $\mathbb{F}_2^s$ uses the fixed basis of $\mathbb{F}_{2^s}$. Fix linear maps

$$A : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^t, \quad C : \mathbb{F}_2^t \rightarrow \mathbb{F}_2^s, \quad C \circ A = 0.$$

Setup samples $\alpha_i \leftarrow \mathbb{F}_{2^s}^\times$ independently for $0 \leq i < R$. Starting from $Q_0 := 0$, the inner computes

$$Y_i := X_i + A(Q_i), \quad Q_{i+1} := \alpha_i Q_i + C(X_i) \quad (0 \leq i < R). \quad (33)$$

It outputs $I(X) := (Y_0, \dots, Y_{R-1})$ and discards Q_R . Field multiplication refreshes a nonzero state once per t output coordinates. The fixed maps A and C perform the cheaper work inside each step. The identity $C \circ A = 0$ supports the transfer analysis and the optimized transpose; it is not needed for injectivity.

Lemma 6.1 (Inner bijectivity). *For every fixed multiplier schedule and every fixed pair of linear maps A, C , the map $I : (\mathbb{F}_2^t)^R \rightarrow (\mathbb{F}_2^t)^R$ defined in (33) is bijective.*

Proof. Fix $Y_0, \dots, Y_{R-1}$. The initial state is known. At step i , recover

$$X_i = Y_i + A(Q_i),$$

and then compute Q_{i+1} from (33). This procedure recovers exactly one input sequence from every output sequence. $\square$

6.6 Composition and rate

Definition 6.2 (Structured SPIN). For a selector θ_m and one realization of its setup, the Structured SPIN encoder is

$$E_m := I_m \circ \Pi_m \circ O_m : \mathbb{F}_2^{K_m} \longrightarrow \mathbb{F}_2^{N_m}. \quad (34)$$

Every realized local map is injective. The outer direct sum is therefore injective. The interleaver is a permutation, and Lemma 6.1 makes the inner bijective. Hence every realized Structured SPIN encoder is injective and has rate

$$\frac{K_m}{N_m} = \frac{k_m}{b_m}. \quad (35)$$

The distance analysis can now vary the constituent family without changing the routing or inner definitions. It must supply a certified outer spectrum or envelope and a conditional failure envelope for the structured route classes used in Theorem 3.1.

The scalable family below lets the base constituent and b_m vary while retaining the fixed RM2Sub step and state dimensions. Finite-length instantiations may tune these choices at one target length. Both cases use Definition 6.2.

6.7 Scalable constituent family

The scalable outer begins with an even base length a and an injective rate-one-half map

$$H_a : \mathbb{F}_2^{a/2} \longrightarrow \mathbb{F}_2^a.$$

Its weight enumerator

$$W_a(z) := \sum_{v \in \mathbb{F}_2^{a/2}} z^{\text{wt}(H_a(v))}$$

must be known exactly or bounded by a certified envelope. For any multiple b of a , let $H_{a,b}$ be the direct sum of b/a copies of H_a . Setup samples two independent permutations $\tau_1, \tau_2 \leftarrow S_b$ and constructs the rate-one-half BA-3 constituent

$$\mathcal{O}_{a,b} := \text{Acc}_b \circ \tau_2 \circ \text{Acc}_b \circ \tau_1 \circ H_{a,b}. \quad (36)$$

It samples $\mathcal{O}_{a,b}$ once and reuses that realization in every outer position. This is the chosen-block form of the BA construction [KPR⁺26, PR26].

The base length a is a construction parameter. A distance certificate must bind a particular H_a , its spectrum, and an admissible schedule for b . Changing a or H_a changes the outer enumerator and may change the certified schedule. It does not change the route or the structured inner.

Current certified choice. For the theorem below, set $a := 24$ and let

$$H_{24} : \mathbb{F}_2^{12} \longrightarrow \mathbb{F}_2^{24}$$

encode the extended binary Golay code. Fix a positive integer m and set $L_m := 128m$. Let b_m be the least positive multiple of 24 satisfying

$$b_m \geq \frac{39}{4} \log_2(L_m b_m), \quad N_m := L_m b_m, \quad K_m := N_m/2. \quad (37)$$

Then $b_m = \frac{39}{4} \log_2 N_m + O(1)$, and consecutive direct lengths have relative gap $o(1)$.

Use $\mathcal{O}_{24, b_m}$ from (36). Setup reuses this sampled map in all L_m diagonal positions:

$$G_{m,1} = \dots = G_{m,L_m} := \mathcal{O}_{24, b_m}.$$

The expectation over (τ_1, τ_2) therefore remains outside the q th enumerator power, as in (31).

After sampling this shared constituent, setup follows the generic construction. It samples one independent $\sigma_j \leftarrow S_{b_m}$ for every outer position and one independent $\pi_r \leftarrow S_{L_m}$ for every region. The inner fixes $(t, s) = (128, 19)$ and the RM2Sub-S19 maps A, C . It samples one independent $\alpha_i \leftarrow \mathbb{F}_{2^{19}}^\times$ per inner step. All route permutations and multipliers are independent of (τ_1, τ_2) .

Theorem 6.3 (Scalable Structured SPIN). *For the setup above,*

$$\mathbb{P}[d_{\min}(E_m) \leq \lfloor 0.11N_m \rfloor] = o(1) \quad (m \rightarrow \infty). \quad (38)$$

The probability includes the two BA permutations, all route permutations, and all RM2Sub multipliers. Every realized code has rate $1/2$. Its ordinary and transposed encoders use $O(N_m)$ bit operations.

Appendix A.1 records the certified outer event, the occupation split, and the outward numerical inequalities used in Theorem 6.3. The block constant $39/4$ is the current safe certificate value. Tightening it changes (37), but not the code architecture.

For any sufficiently large requested output length n , let $m(n)$ be the largest index satisfying $N_{m(n)} \leq n$. Define

$$\tilde{E}_n(x) := E_{m(n)}(x) \| 0^{n-N_{m(n)}}. \quad (39)$$

This zero extension preserves injectivity and minimum distance. Consecutive direct lengths have relative gap $o(1)$, so $N_{m(n)}/n = 1 - o(1)$. The wrapped family therefore has rate $1/2 - o(1)$, relative distance greater than $0.11 - o(1)$ with high probability, and $O(n)$ ordinary and transposed work. The native-length theorem remains the primary statement.

6.8 Finite-length specializations

At a fixed target length, Definition 6.2 permits hand-tuned outer and routing parameters. In particular, the base length a and base constituent H_a need not match the current Golay certificate. The current $N = 2^{21}$ evaluation target retains RM2Sub-S19, but its outer constituent remains under selection. Appendix A.2 records the open certification and evaluation boundary for this specialization.

7 Finite-length certificates

The first-moment identity is finite and exact. A computer-assisted proof upper-bounds its class contributions and should expose a finite collection of rigorous bounds rather than only a floating optimization trace.

Fix a finite class map with class set $\mathcal{T}$. A certificate provides bounds $\hat{A}_\tau^{\text{out}}$ and $\hat{Q}_\tau(d)$ satisfying

$$A_\tau^{\text{out}} \leq \hat{A}_\tau^{\text{out}}, \quad Q_\tau(d) \leq \hat{Q}_\tau(d)$$

for every τ . If an independently checkable evaluation proves

$$\sum_{\tau \in \mathcal{T}} \hat{A}_\tau^{\text{out}} \hat{Q}_\tau(d) < 1, \tag{40}$$

then Theorem 3.1 proves the existence of an injective encoder with minimum distance greater than d . The bound $\hat{Q}_\tau(d)$ may certify a worst-case conditional failure probability; the conditional law need not be identical throughout the class.

A certificate must record:

1. the complete encoder and setup distribution;
2. all finite parameters and divisibility conditions;
3. the interface classes and their complete range;
4. the source of each exact value or upper bound;
5. the arithmetic and outward-rounding rules;
6. the verifier and hashes of its inputs.

The main text reports the class partition and final bound. Detailed tables, optimization grids, and verifier transcripts belong in the artifact bundle or appendix. Appendix A.1 identifies the hash-bound bundle for the scalable Structured SPIN theorem. Future finite specializations should use the same manifest fields and evidence labels.

8 Finite lengths, asymptotics, and complexity

Finite validity, asymptotic distance, and encoding complexity are separate properties. We state them separately for each SPIN family.

8.1 Native and requested lengths

The direct constructions apply at native lengths for which their block dimensions are integral. Their logarithmic schedules permit admissible lengths with relative gaps $o(1)$. For Structured SPIN, the zero extension in (39) maps the largest native member below any sufficiently large requested length into that output length. This wrapper preserves injectivity and absolute minimum distance. It changes rate and relative distance by only $o(1)$ and preserves linear work.

8.2 Asymptotic statements

Theorems 4.5 and 5.2 give positive relative distance with high probability for the independent-block ensembles. They specify the block schedule, setup failure probability, and rate. At rate one half, the stated examples use block constants 3 and 17, respectively; Random SPIN also uses memory constant 51/50.

Theorem 6.3 uses direct lengths $N_m = L_m b_m$, where $L_m = 128m$ and b_m is the least multiple of 24 satisfying $b_m \geq \frac{39}{4} \log_2 N_m$. These lengths have relative gaps $o(1)$. The requested-length wrapper gives rate $1/2 - o(1)$, relative distance $0.11 - o(1)$, and linear work.

8.3 Random-family encoding costs

Under either proved schedule, the outer encoder processes N/B blocks of length $B = \Theta(\log N)$. Direct dense block multiplication therefore costs $\Theta(N \log N)$ bit operations. The accumulator costs $O(N)$. The Random SPIN recurrence evaluates $m = \Theta(\log N)$ taps per output coordinate and hence costs $\Theta(N \log N)$. These bounds concern direct encoders; preprocessing or word-level batching may reduce practical constants.

8.4 Cost model

Let $C_{\text{out}}(b)$ be the work to encode one outer block, and let $C_{\text{in}}(t, s)$ be the work for one structured inner step. A Structured SPIN encoder has the schematic cost

$$T(N) = \frac{N}{b} C_{\text{out}}(b) + \frac{N}{t} C_{\text{in}}(t, s) + O(N), \quad (41)$$

where the last term covers routing and permutation work.

For any fixed base constituent with a constant-size encoding circuit, tiling, two permutations, and two accumulators give $C_{\text{out}}(b) = O(b)$. The fixed RM2Sub-S19 step has constant cost. Every term in (41) is therefore $O(N)$. The current certificate uses extended Golay as the base constituent, but the cost argument does not depend on its 24-bit length.

9 Implementation and performance

The mathematical sections define forward encoders. The implementation uses transposed, batched evaluation because the target application encodes many binary columns in parallel. This evaluation order must preserve the same linear map.

The current finite-length target uses a BA-3 outer, the factored Structured SPIN interleaver, and the RM2Sub-S19 inner. The base constituent and its length remain under selection. The optimized transposed pipeline applies the transposed inner, inverts the routing factors, and then applies the transposed BA-3 outer.

Performance measurements and distance certificates answer different questions. Correctness tests establish that the optimized code evaluates the declared linear map. Benchmarks measure that evaluation. The finite first-moment certificate separately establishes minimum distance.

TODO. Select the finite BA-3 base constituent and insert the finalized interface, build and test commands, operation ledger, dependency provenance, correctness receipt, and serialized benchmark receipt. Do not report a preliminary timing as a paper result before these records are fixed.

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A Structured SPIN certificate and finite specialization

A.1 Certificate for the scalable family

This subsection proves Theorem 6.3. The proof first selects one Golay–BA-3 constituent whose realized spectrum obeys a certified envelope. It then reduces the structured route to occupation and mean row weight. The final first-moment bound separates four regimes: $Q = 1, 2$, fixed $Q \geq 3$, growing sparse Q , and positive relative occupation. Finite RM2Sub transfers handle the last two regimes. Continuum transfers handle the fixed regimes.

For $0 \leq d \leq N$ and $1 \leq Q \leq L$, let $Z_{d,Q}$ count nonzero messages with exactly Q active outer blocks and encoded weight at most d . Unless a display states otherwise, expectations and probabilities include the shared BA constituent, the route permutations, and the RM2Sub multipliers. The finite-length specialization is not part of this probability space.

A.1.1 The one-sampled Golay–BA-3 outer

We begin with the exact expected spectrum before selecting the shared constituent. Let

$$G(u) := 1 + 759u^8 + 2576u^{12} + 759u^{16} + u^{24}$$

be the weight polynomial of the extended Golay code, and define

$$G_b(a) := [u^a]G(u)^{b/24}.$$

For $a \geq 1$, put $r_a := \lceil a/2 \rceil$ and

$$T_b(a, c) := \binom{c-1}{r_a-1} \binom{b-c}{a-r_a}, \quad P_b(a, c) := \frac{T_b(a, c)}{\binom{b}{a}}. \quad (42)$$

We set $T_b(0, 0) := 1$ and $T_b(0, c) := 0$ for $c \neq 0$. Binomial coefficients outside their natural range are zero. Thus $P_b(a, c)$ is the exact weight-transition probability of a uniformly interleaved accumulator.

For a realized BA constituent $\mathcal{O}_b$, let $A_{\mathcal{O}_b}(w)$ count its nonzero words of weight w , and define

$$\bar{A}_b(w) := \mathbb{E}_{\tau_1, \tau_2}[A_{\mathcal{O}_b}(w)].$$

Lemma A.1 (Exact expected BA spectrum). *For every $0 \leq w \leq b$,*

$$\bar{A}_b(w) = \sum_{a=0}^b \sum_{c=0}^b G_b(a) P_b(a, c) P_b(c, w). \quad (43)$$

Proof. Fix a Golay direct-sum word of weight a . The first uniform permutation and accumulator produce weight c with probability $P_b(a, c)$. Conditioned on that weight, the second permutation makes the word uniform on its Hamming slice. The second accumulator then produces weight w with probability $P_b(c, w)$. Summing over the $G_b(a)$ input words proves (43). $\square$

The exact formula supplies both the tail bound and the spectrum envelope. Let

$$\mathcal{W} := [0.104, 0.896].$$

The outer certificate proves

$$\sum_{w/b \notin \mathcal{W}} \bar{A}_b(w) = o(1). \quad (44)$$

Lemma A.2 (One-sample selection). *There is an event $\mathcal{G}_b$, determined by the two BA permutations, such that*

$$\mathbb{P}[\mathcal{G}_b] = 1 - o(1). \quad (45)$$

On $\mathcal{G}_b$, every nonzero constituent word has relative weight in $\mathcal{W}$, and

$$A_{\mathcal{O}_b}(w) \leq b^2 \bar{A}_b(w) \quad (0 \leq w \leq b). \quad (46)$$

Proof. The expected number of nonzero tail words equals the left side of (44). Markov's inequality shows that no such word exists with probability $1 - o(1)$. For each remaining weight w , another application of Markov's inequality gives

$$\mathbb{P}[A_{\mathcal{O}_b}(w) > b^2 \bar{A}_b(w)] \leq b^{-2}.$$

A union bound over at most $b + 1$ weights costs $O(1/b)$. Intersecting the two events proves the lemma. $\square$

The event in Lemma A.2 is paid once. After setup samples $\mathcal{O}_b$, the construction reuses that realization at every outer position.

We next state the asymptotic envelope verified from (43). Let

$$h(x) := -x \ln x - (1-x) \ln(1-x)$$

with its continuous endpoint values. Define

$$g(a) := \inf_{u>0} \left\{ \frac{1}{24} \ln G(u) - a \ln u \right\}.$$

On the accumulator feasibility region $a/2 \leq c \leq 1 - a/2$, set

$$\pi(a, c) := ch\left(\frac{a}{2c}\right) + (1 - c)h\left(\frac{a}{2(1 - c)}\right) - h(a),$$

and set $\pi(a, c) := -\infty$ outside that region. The limiting exponent is

$$a_{\text{BA}}(x) := \sup_{0 \leq a, c \leq 1} \{g(a) + \pi(a, c) + \pi(c, x)\}. \quad (47)$$

Uniform coefficient and Stirling bounds in (43) give

$$\frac{1}{b} \ln \bar{A}_b(w) \leq a_{\text{BA}}(w/b) + O\left(\frac{\ln b}{b}\right). \quad (48)$$

The certificate represents a concave majorant $\hat{a}_{\text{BA}}$ as the lower envelope of rational affine supports. It proves

$$a_{\text{BA}}(x) \leq \hat{a}_{\text{BA}}(x) \quad (x \in \mathcal{W}). \quad (49)$$

There are 31 affine segments. The central support is the exact dimension bound $\hat{a}_{\text{BA}}(x) \leq (\ln 2)/2$. The interval verifier processes 101,910 rational boxes. Its largest accepted outward upper endpoint is

$$-2.2000492744786192 \cdot 10^{-9}. \quad (50)$$

Equations (48) and (49) give the outer exponent used below. The artifact retains all affine supports and box witnesses.

A.1.2 What the structured route preserves

View an outer word as an L -tuple of rows in $\mathbb{F}_2^b$. For the framework in Section 3, take its class to be the ordered row-weight profile

$$c(u) := (\text{wt}(u_1), \dots, \text{wt}(u_L)).$$

This finite class records the information preserved by the independent block shuffles. The analysis below constructs a uniform route-inner failure envelope for every such profile. It then aggregates profiles by occupation and mean row weight where those coarser parameters suffice.

Fix Q active outer positions. Let their realized output weights be $w_1, \dots, w_Q$, and set

$$x_j := \frac{w_j}{b}, \quad \alpha := \frac{Q}{L}, \quad x := \frac{1}{Q} \sum_{j=1}^Q x_j, \quad q := \alpha x.$$

The next lemma justifies the reduction from the complete weight tuple to (α, x) when α has a positive lower bound.

Lemma A.3 (Route domination). *Let F be any nonnegative function of the routed L -by- b binary array. Conditioned on the row weights $w_1, \dots, w_Q$, the structured route satisfies*

$$\mathbb{E}_{\text{route}}[F] \leq (b + 1)^Q (L + 1)^b \mathbb{E}_{\text{Ber}(q)^{Lb}}[F]. \quad (51)$$

If α is bounded below by a positive constant, the multiplicative factor in (51) is $\exp(o(N))$.

Proof. For row j , begin with b independent Bernoulli- x_j variables and condition their sum to equal w_j . The value w_j is a mode of the binomial distribution. Hence the inverse conditioning probability is at most $b + 1$. Applying this reduction to all active rows costs at most $(b + 1)^Q$.

Before the region shuffle, one fixed region contains independent Bernoulli variables whose average success probability is q . Let S be their sum. For every $0 \leq k \leq L$, the Chernoff bound for a Poisson-binomial sum gives

$$\mathbb{P}[S = k] \leq \exp\{-LD_{\text{KL}}(k/L\|q)\}.$$

The method-of-types lower bound gives

$$\mathbb{P}[\text{Bin}(L, q) = k] \geq \frac{1}{L+1} \exp\{-LD_{\text{KL}}(k/L\|q)\}.$$

After a uniform region shuffle, both laws are uniform on every weight- k slice. The routed region law is therefore pointwise at most $L + 1$ times the iid Bernoulli- q law. Multiplying over the b independent region shuffles proves (51).

Finally,

$$\ln((b+1)^Q(L+1)^b) = Q \ln(b+1) + b \ln(L+1) = o(Lb)$$

when Q/L is bounded below by a positive constant and $b, L \rightarrow \infty$. $\square$

Concavity of $\widehat{a}_{\text{BA}}$ now reduces the selected outer spectrum to the mean row weight:

$$\frac{1}{Q} \sum_{j=1}^Q \widehat{a}_{\text{BA}}(x_j) \leq \widehat{a}_{\text{BA}}(x). \quad (52)$$

The region shuffles are essential for Lemma A.3. The deterministic transpose alone spreads each row, but it does not erase the row's coordinate pattern.

A.1.3 A finite RM2Sub transfer

We next define the transfer matrix used after the route reduction. Identify the RM2Sub state field with $\mathbb{F}_2^{19}$, and set

$$n := 2^{19}, \quad M := n - 1, \quad \kappa := \frac{M}{M - 1}.$$

The fixed state-output map A has image weights

$$(d_i)_{i=0}^6 = (0, 48, 56, 64, 72, 80, 128)$$

with multiplicities

$$(c_i)_{i=0}^6 = (1, 5040, 110848, 292510, 110848, 5040, 1).$$

For $\xi, \eta \in \mathbb{F}_2^{19}$, define the integer association matrix

$$K_{ij} := \sum_{\xi: \text{wt}(A(\xi))=d_i} \sum_{\eta: \text{wt}(A(\eta))=d_j} (-1)^{\langle \xi, \eta \rangle}. \quad (53)$$

The frozen circuit and an integer Walsh transform determine K exactly.

Fix an iid input-bit probability $\beta \in (0, 1)$ and a weight fugacity $z \in (0, 1)$. Put

$$u := 1 - \beta + \beta z, \quad v := \beta + (1 - \beta)z, \quad r := 1 - \beta - \beta z,$$

and define vectors $\mathbf{D}, \mathbf{F} \in \mathbb{R}^7$ by

$$D_i := u^{128-d_i} v^{d_i}, \quad F_i := u^{128-d_i} r^{d_i}.$$

Set

$$k_{00} := \frac{1}{n} \sum_i c_i F_i, \quad k_{01} := u^{128} - k_{00}, \quad (54)$$

$$\bar{\mu} := \frac{n^{-1} \mathbf{F}^\top \mathbf{K} \mathbf{D} - k_{00} D_0}{k_{01}}, \quad \mu := \frac{\sum_i c_i D_i - D_0}{M}. \quad (55)$$

The three state classes are zero (Z), freshly produced nonzero (D), and uniform or punctured-uniform nonzero (R). Define

$$T_3(\beta, z) := \begin{pmatrix} k_{00} & k_{01} & 0 \\ \bar{\mu}/M & 0 & \bar{\mu} \\ \kappa\mu/M & 0 & \kappa\mu \end{pmatrix}. \quad (56)$$

Let $e_Z := (1, 0, 0)^\top$, and let $\mathbf{1}_3$ be the all-ones vector in $\mathbb{R}^3$.

Lemma A.4 (Three-state RM2Sub envelope). *For iid Bernoulli- β inner input and zero initial state,*

$$\mathbb{E}[z^{\text{wt}(Y)}] \leq e_Z^\top T_3(\beta, z)^{N/128} \mathbf{1}_3. \quad (57)$$

The expectation is over the iid Bernoulli input bits and the independent nonzero field multiplier in every inner step.

Proof. The first row of (56) is exact. Equation (53) evaluates the output moment after a transition from zero to a nonzero state, which gives $\bar{\mu}$. For a uniform nonzero entering state, the image spectrum gives μ .

For any fixed nonzero entering state, the fresh multiplier makes the multiplied state uniform on $\mathbb{F}_{2^{19}}^\times$. Adding a nonzero syndrome terminates the state with probability $1/M$. Conditioned on survival, the next-state law omits at most one nonzero value. Replacing that law by a uniform nonzero law costs at most $\kappa = M/(M-1)$. The last row pays this factor. All entries are nonnegative, so transfer multiplication proves (57). $\square$

A.1.4 Positive occupation

Fix $\alpha \geq 10^{-4}$ and $x \in \mathcal{W}$. We compare the routed Bernoulli- q input, where $q = \alpha x$, with a reference bit probability $\beta = py$. The binary relative-entropy chain rule gives

$$D_{\text{KL}}(\alpha x \| py) \leq D_{\text{KL}}(\alpha \| p) + \alpha D_{\text{KL}}(x \| y). \quad (58)$$

For a nonnegative matrix T , let $\rho_{\text{sp}}(T)$ denote its spectral radius. A Collatz witness for T is a positive vector ω and a scalar λ satisfying $T\omega \leq \lambda\omega$ entrywise; it certifies $\rho_{\text{sp}}(T) \leq \lambda$. For $p, y, z \in (0, 1)$, define

$$\begin{aligned} \Phi(\alpha, x; p, y, z) &:= \alpha \hat{a}_{\text{BA}}(x) + D_{\text{KL}}(\alpha \| p) + \alpha D_{\text{KL}}(x \| y) \\ &\quad + \frac{1}{128} \ln \rho_{\text{sp}}(T_3(py, z)) - 0.11 \ln z. \end{aligned} \quad (59)$$

Proposition A.5 (Positive-occupation certificate). *For every $(\alpha, x) \in [10^{-4}, 1] \times \mathcal{W}$, there exist rational $p, y, z \in (0, 1)$ and a rational positive Collatz vector such that the outward upper bound on (59) is negative. The largest accepted upper endpoint is*

$$-3.194048334767875 \cdot 10^{-7}. \quad (60)$$

Proof. Condition on $\mathcal{G}_b$ and a fixed set of $Q = \alpha L$ active rows. Equations (46) and (49) bound each row-weight multiplicity. Lemma A.3 replaces the structured route by iid Bernoulli- αx input at subexponential cost. Equation (52) combines unequal row weights.

Changing the reference bit probability to py costs the exponent in (58). Lemma A.4 gives the RM2Sub moment. The Chernoff inequality contributes $-0.11 \ln z$ per output coordinate. These terms give (59).

The verifier partitions the (α, x) rectangle into 621 rational boxes. It fixes one witness (p, y, z) and one Collatz vector before checking the vertices of each box. For one affine support of $\widehat{a}_{\text{BA}}$, the exponent is convex in $(\alpha, q = \alpha x)$. Four vertex checks therefore cover the box. All logarithms and matrix operations use outward interval arithmetic. The largest endpoint is (60). $\square$

The quantifier order in Proposition A.5 is

$$\sup_{\alpha, x} \inf_{p, y, z}.$$

The witnesses are proof parameters. They are not setup randomness.

A.1.5 Fixed occupation

The subexponential estimate in Lemma A.3 is not uniform when Q is fixed. We instead retain one coefficient fugacity for each active row.

Set

$$r_0 := \frac{1}{M}, \quad p_0 := \frac{2^{18}}{M}, \quad P_0 := \begin{pmatrix} 0 & 1 \\ r_0 & 1 - r_0 \end{pmatrix}.$$

For $\theta > 0$, let $z_L := e^{-\theta/L}$ and $\gamma := p_0 \theta$. Define

$$D_\gamma(t) := \text{diag}(1, e^{-\gamma t}).$$

If one region contains a impulses, its normalized locations converge to the order statistics of a uniform points. Their spacings are uniform on the a -simplex. Define the limiting region matrix

$$K_a(\theta) := a! \int_{\substack{t_i \geq 0 \\ \sum_{i=0}^a t_i = 1}} D_\gamma(t_0) P_0 D_\gamma(t_1) \cdots P_0 D_\gamma(t_a) d\mathbf{t}. \quad (61)$$

For $a = 0$, this means $K_0(\theta) := D_\gamma(1)$.

For fixed Q , collision and boundary events change every nonzero entry by a relative factor $1 + O_Q(M/L)$. Since $b = O(\log L)$, their total logarithmic cost across all regions is $o_Q(b)$. Thus (61) has the same exponential rate as the exact full-state region transfer.

Introduce a row fugacity $u_j > 0$ for each active row and define

$$T_Q(\mathbf{u}, \theta) := \sum_{\mathbf{e} \in \{0,1\}^Q} \left(\prod_{j=1}^Q u_j^{e_j} \right) K_{\sum_j e_j}(\theta). \quad (62)$$

Coefficient extraction enforces the row weights. For a fixed weight tuple,

$$\mathbb{E}[z_L^{\text{wt}(Y)} \mid w_1, \dots, w_Q] \leq \left(\prod_{j=1}^Q \binom{b}{w_j}^{-1} u_j^{-w_j} \right) \cdot e_0^\top T_Q(\mathbf{u}, \theta)^b \mathbf{1}_2 \exp(o_Q(b)). \quad (63)$$

Here $e_0 := (1, 0)^\top$ and $\mathbf{1}_2 := (1, 1)^\top$. The expectation is over the route permutations and RM2Sub multipliers, conditioned on the displayed row weights and the selected constituent.

We use one product norm for every fixed $Q \geq 3$. Fix $0 < \sigma < 1$, $\tau > 0$, and $v_1 > 0$. Let

$$H_u := I + uP_0, \quad \mathbf{v} := (1, v_1)^\top.$$

For a nonnegative 2-by-2 matrix B , define

$$\|B\|_{\mathbf{v}} := \max \left\{ (B\mathbf{v})_0, \frac{(B\mathbf{v})_1}{v_1} \right\}.$$

Also define

$$G_Q(\sigma, \tau) := \sup_{0 \leq y \leq 1} \left\{ -\tau y + \left(1 + \frac{1}{Q} \right) \ln(1 - y + y/\sigma) \right\}, \quad (64)$$

$$m_{\mathbf{v}}(u) := \max \left\{ 1 + u\sigma v_1, \frac{ur_0}{v_1} + \sigma(1 + u(1 - r_0)) \right\}. \quad (65)$$

The beta-integral representation of (61), followed by the weighted row norm induced by $\mathbf{v}$, gives

$$\|T_Q(\mathbf{u}, Q\tau/p_0)\|_{\mathbf{v}} \leq \exp\{QG_Q(\sigma, \tau)\} \prod_{j=1}^Q m_{\mathbf{v}}(u_j). \quad (66)$$

The logarithm in (64) is nonnegative. Hence $G_Q(\sigma, \tau) \leq G_3(\sigma, \tau)$ for every $Q \geq 3$.

For $x \in \mathcal{W}$, define the coupled row exponent

$$R(x; u) := \widehat{a}_{\text{BA}}(x) - h(x) - x \ln u + \ln m_{\mathbf{v}}(u). \quad (67)$$

Combining (46), (63), and (66) yields

$$\begin{aligned} \ln \mathbb{E}[Z_{\lfloor 0.11N \rfloor, Q} \mid \mathcal{G}_b] &\leq Q \ln L \\ &\quad + b \sum_{j=1}^Q \left\{ R(x_j; u_j) + G_3(\sigma, \tau) + \frac{0.11\tau}{p_0} \right\} + o_Q(b). \end{aligned} \quad (68)$$

Proposition A.6 (Weight-coupled fixed occupation). *Fix any integer $Q \geq 3$. Under (37),*

$$\mathbb{P}[Z_{\lfloor 0.11N \rfloor, Q} > 0 \mid \mathcal{G}_b] \leq \exp\{-0.0008689 Qb + o_Q(b)\}. \quad (69)$$

Proof. The certificate fixes

$$\sigma = \frac{127}{250}, \quad \tau = \frac{133}{125}, \quad v_1 = \frac{3}{1600}.$$

For each of the 31 affine segments of $\widehat{a}_{\text{BA}}$, it chooses one positive rational fugacity u . For fixed u , the map $x \mapsto R(x; u)$ is convex on that segment. It is therefore enough to check the two endpoints. The resulting 62 outward checks prove

$$R(x; u) + G_3(\sigma, \tau) + \frac{0.11\tau}{p_0} < -0.07196093457399053. \quad (70)$$

Under $b = (39/4) \log_2 N + O(1)$, choosing an active position costs at most

$$\frac{4 \ln 2}{39} b + o(b).$$

The largest outward endpoint after adding this cost is -0.0008689160550218037 . Equation (68) and Markov's inequality prove (69). The factor b^2 in (46) contributes only $o_Q(b)$. $\square$

Proposition A.7 (One- and two-row occupations). *Under (37), the bad-message probabilities for $Q = 1$ and $Q = 2$ satisfy*

$$\mathbb{P}[Z_{\lfloor 0.11N \rfloor, Q} > 0 \mid \mathcal{G}_b] = o(1).$$

Proof. The exact continuum transfers for $Q = 1, 2$ give a structured-outer gap greater than 0.2592968543039901 bits per active row. After the support choice, the schedule-decay exponent is greater than 1.5281443294639036 per active row. These margins leave a strictly negative logarithmic first-moment exponent after all counting costs. The hash-bound receipt verifies the stated outward endpoints. $\square$

A.1.6 Growing sparse occupation

It remains to cover $Q \rightarrow \infty$ with $Q/L \leq 10^{-4}$. In this regime, a four-state transfer avoids paying the punctured-uniform factor κ at every step. Its state classes are zero (Z), freshly produced nonzero (D), uniform nonzero (U), and punctured-uniform nonzero (P).

Let p be a mark probability and set the input-bit probability to $\beta = p/2$ in (54) and (55). Define

$$a(p) := 1 - \frac{1}{n} \sum_{i=0}^6 c_i (1-p)^{d_i}.$$

For the parameter range used below, the following matrix is an entrywise envelope:

$$T_4(p, z) := \begin{pmatrix} k_{00} & k_{01} & 0 & 0 \\ a/M & 0 & \bar{\mu} - z^{128}a & a(M-1)/M \\ a/M & 0 & \mu - z^{128}a & a(M-1)/M \\ \kappa a/M & 0 & \kappa(\mu - z^{128}a) & \kappa a(M-1)/M \end{pmatrix}, \quad (71)$$

where $a = a(p)$. The fresh multiplier makes termination independent of the current output after the state and input are fixed. This gives the factors $1/M$ and $(M-1)/M$ in (71).

Set

$$p := \frac{8}{5}\alpha, \quad z := 1 - \frac{8}{5}\alpha, \quad \mathbf{w} := (1, 2^{-10}, 2^{-10}, \kappa 2^{-10})^\top.$$

Exact rational Bernstein coefficients prove

$$T_4(p, z)\mathbf{w} \leq (1 - 99\alpha)\mathbf{w} \quad (0 < \alpha \leq 10^{-4}). \quad (72)$$

The induced random-outer exponent satisfies

$$\begin{aligned} \frac{\alpha \ln 2}{2} + D_{\text{KL}}(\alpha \| p) + \frac{1}{128} \ln(1 - 99\alpha) - 0.11 \ln z \\ \leq -\frac{780897}{6665600} \alpha. \end{aligned} \quad (73)$$

The certificate proves (72) after clearing all positive denominators. The three nontrivial row polynomials have degrees at most 511. After their zero factors are removed, every Bernstein coefficient is strictly negative.

To transfer (73) to the selected BA outer, define its likelihood excess relative to a random rate-one-half row:

$$\ell(x) := \widehat{a}_{\text{BA}}(x) - h(x) + \frac{\ln 2}{2}. \quad (74)$$

On each affine segment of $\widehat{a}_{\text{BA}}$, the function ℓ is convex. Outward endpoint checks prove

$$\sup_{x \in \mathcal{W}} \ell(x) < 0.01281. \quad (75)$$

Thus each active selected-outer row costs at most $\exp((0.01281 + o(1))b)$ relative to the random-outer counting measure.

Proposition A.8 (Growing sparse occupation). *Suppose $Q \rightarrow \infty$ and $Q/L \leq 10^{-4}$. Then*

$$\mathbb{E}[Z_{\lfloor 0.11N \rfloor, Q} \mid \mathcal{G}_b] \leq \exp\{-(0.0325484 + o(1))Qb\}. \quad (76)$$

Proof. Equations (73) and (75) leave the natural decay coefficient

$$\kappa_{\text{BA}} := \frac{780897}{6665600} - 0.01281 = \frac{86938833}{833200000} > 0.10434329$$

per active row bit. Choosing the Q active rows costs at most $(4 \ln 2/39 + o(1))Qb$. The exact bound $\ln 2 < 7/10$ gives

$$\kappa_{\text{BA}} - \frac{4 \ln 2}{39} > 0.0325484.$$

The b^2 selection factor and the local-limit remainders contribute $o(Qb)$ because $Q \rightarrow \infty$. This proves (76). $\square$

A.1.7 Completion and certificate boundary

Proof of Theorem 6.3. Condition on $\mathcal{G}_b$, and abbreviate $Z_Q := Z_{\lfloor 0.11N \rfloor, Q}$. Proposition A.7 covers $Q = 1, 2$, and Proposition A.6 covers every fixed $Q \geq 3$. Proposition A.8 covers $Q \rightarrow \infty$ with $Q/L \leq 10^{-4}$. Proposition A.5 covers $Q/L \geq 10^{-4}$.

These cases cover every sequence of occupations. The fixed bounds tend to zero for each fixed Q . The growing sparse bound is geometric on the Qb scale, even after summing over the possible values of Q . The positive-range bound is uniform and exponential on the N scale. Markov's inequality and a union bound therefore give

$$\mathbb{P} \left[\sum_{Q=1}^L Z_Q > 0 \mid \mathcal{G}_b \right] = o(1).$$

This is (38) conditioned on $\mathcal{G}_b$. Equation (45) removes the conditioning.

Rate follows from (35). Each outer encoder uses constant-size Golay circuits, two stored permutations, and two accumulators, so it costs $O(b)$ in either circuit direction. The route and the fixed RM2Sub steps cost $O(N)$. Thus the ordinary and transposed encoders both use $O(N)$ bit operations. $\square$

The paper-owned certificate bundle is stored in the `single_sampled_ba_rm2sub` directory under `workstreams/paper_architecture/certificates`. Its README records the upstream manifest filename and SHA-256 hash. The manifest binds 23 theory files, five companion linear-time files, and three frozen dependencies. The relocated-bundle checker verifies all 31 entries.

The appendix and artifact deliberately separate analytic obligations from finite computation. The appendix proves the enumerator identities, route domination, transfer reductions, convexity steps, and occupation split. The artifact stores rational affine supports, box covers, Collatz vectors, Bernstein coefficients, and outward endpoints. The present theorem relies on the hash-bound Python and interval-arithmetic certificate.

A.2 Finite-length specialization

The finite-length target at $N = 2^{21}$ will use a BA-3 outer and the RM2Sub-S19 inner. Its base constituent, base length, outer-block length, and routing parameters remain under selection. The finite design will optimize measured cost subject to an exact first-moment certificate. The scalable theorem does not depend on this optimization.

TODO. Insert the selected finite parameters and exact setup law. Certify the finite BA-3 spectrum envelope, evaluate the complete first-moment sum with outward-rounded arithmetic, and bind the resulting theorem inputs to the finalized implementation manifest before stating a finite-distance theorem.

B Proof details and artifact interfaces

This appendix records proof details that are not needed to understand the construction progression.

B.1 Random SPIN exponent closure

This subsection proves the two estimates used by Theorem 5.2. The outer exponent follows from the exact injection law in (16). For a sequence $h/N \rightarrow \eta \in (0, 1)$, coefficient extraction gives

$$\Phi_{\text{blk}}(\eta) = \begin{cases} \eta \log_2(1 + \sqrt{2}), & 0 < \eta \leq 1 - 2^{-1/2}, \\ H_2(\eta) - 1/2, & 1 - 2^{-1/2} \leq \eta < 1. \end{cases} \quad (77)$$

The inner exponent comes from the transfer matrix in (20). For $a > 0$ and $S \in (0, 1)$, define

$$\lambda_{\infty}(a, S) := \max \left\{ 1, \frac{(1+a)(1+S)}{2} \right\}.$$

Lemma B.1 (Limiting convolution growth). *If $m_N = \Theta(\log N)$, then*

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \log_2 \left(e_{m_N}^{\top} T_{m_N}(a, S)^N \mathbf{1} \right) \leq \log_2 \lambda_{\infty}(a, S). \quad (78)$$

Proof. Set $b := (1 + a)/2$ and let λ_m be the Perron root of $T_m(a, S)$. The right Perron-vector equations give

$$1 = \frac{bS}{\lambda_m} \frac{1 - (b/\lambda_m)^m}{1 - b/\lambda_m} + \left(\frac{b}{\lambda_m}\right)^m \frac{aS}{\lambda_m - 1}. \quad (79)$$

If $b(1 + S) < 1$, this identity gives $\lambda_m \rightarrow 1$. If $b(1 + S) > 1$, the active-state principal submatrix and (79) give $\lambda_m \rightarrow b(1 + S)$. At equality, any subsequential limit greater than one contradicts (79).

Normalize the Perron vector at state zero. The same equations show that the ratio of its largest and smallest coordinates is $\exp(O(m))$. Comparing the all-ones vector with coordinatewise multiples of this Perron vector bounds the matrix power by $\lambda_m^N \exp(O(m))$. Since $m_N = O(\log N)$, this factor is subexponential and proves (78). $\square$

For $\delta \in (0, 1/2)$, equations (22) and (78) give the inner exponent

$$F_\delta(\eta) := \inf_{\substack{a > 0 \\ 0 < S < 1}} [\log_2 \lambda_\infty(a, S) - \eta \log_2 a - \delta \log_2 S - H_2(\eta)]. \quad (80)$$

Set

$$\eta_{\text{conv}}(\delta) := 1 - \frac{1}{2(1 - \delta)}, \quad a_\eta := \frac{\eta}{\sqrt{\eta^2 + \delta^2} + \delta}, \quad S_\eta := \frac{1 - a_\eta}{1 + a_\eta}.$$

Direct minimization yields

$$F_\delta(\eta) = \begin{cases} -\eta \log_2 a_\eta - \delta \log_2 S_\eta - H_2(\eta), & 0 < \eta \leq \eta_{\text{conv}}(\delta), \\ H_2(\delta) - 1, & \eta_{\text{conv}}(\delta) \leq \eta < 1. \end{cases} \quad (81)$$

Lemma B.2 (Linear-weight closure). *Set $\delta_* := 5501/50000$ and $J_{\delta_*} := \Phi_{\text{blk}} + F_{\delta_*}$. Then*

$$J_{\delta_*}(\eta) < 0 \quad (0 < \eta < 1).$$

The inequality is uniform on every compact subinterval of $(0, 1)$.

Proof. Set $\eta_{\text{blk}} := 1 - 2^{-1/2}$. On $(0, \eta_{\text{blk}}]$, the derivative of J_{δ_*} is increasing, so J_{δ_*} is convex and $J_{\delta_*}(0) = 0$. Outward interval arithmetic gives

$$J_{\delta_*}(\eta_{\text{blk}}) \in [-0.074030, -0.074029].$$

Convexity proves strict negativity on the first region. On $[\eta_{\text{blk}}, \eta_{\text{conv}}(\delta_*)]$, the entropy terms cancel and the remaining expression is increasing. Its right endpoint lies in $[-0.011076, -0.011075]$. On the final region,

$$J_{\delta_*}(\eta) = H_2(\eta) + H_2(\delta_*) - \frac{3}{2} \leq H_2(\delta_*) - \frac{1}{2},$$

and the last value lies in $[-0.000023719, -0.000023718]$. These strict intervals prove the claim. $\square$

The remaining issue is uniformity as $\eta \downarrow 0$. Define

$$\eta_* := \frac{1}{5000}, \quad \chi := \frac{101}{100}, \quad \gamma := \frac{51}{50}.$$

Set

$$\begin{aligned} a_* &:= \frac{\eta_*}{\delta_*(\chi + 1)}, & b_* &:= \frac{1 + a_*}{2}, \\ \epsilon_* &:= \frac{\chi^2 a_*}{(\chi + 1)(1 - \chi a_*)} + \frac{\eta_*}{2(1 - \eta_*)}, & \rho_* &:= \delta_*(\chi + 1)e^{\epsilon_*}. \end{aligned}$$

Lemma B.3 (Sparse-weight closure). *Let $m_N := \lceil \gamma \log_2 N \rceil$. There are constants C, N_0 such that, for $N \geq N_0$ and $1 \leq h \leq \eta_* N$,*

$$p_{N,h}(\delta_*) \leq CN^{2\gamma} \sqrt{h} \rho_*^h. \quad (82)$$

If B_N is even, divides N , and satisfies $B_N \geq 17 \log_2 N$, then

$$\sum_{h=1}^{\lfloor \eta_* N \rfloor} \mathbb{E}[A_{N,h}^{\text{out}}] p_{N,h}(\delta_*) = o(1). \quad (83)$$

Proof. For $\eta := h/N$, evaluate (22) at

$$a := \frac{\eta}{\delta_*(\chi + 1)}, \quad S := \frac{1 - \chi a}{1 + a}.$$

The active-state row sum equals $1 - (\chi - 1)a/2 < 1$. The Perron-root identity then bounds the matrix power by $CN^{2\gamma}$ uniformly over the stated range. A standard lower bound on $\binom{N}{h}$ gives (82); the exponent correction increases with η and is at most ϵ_* . Interval arithmetic certifies

$$\rho_* \in [0.2212639483446276, 0.2212639483446278], \\ \gamma \log_2(1/b_*) - 1 > 0.0186697.$$

In particular, $\rho_* < \sqrt{2} - 1$.

The exact independent-block outer law gives

$$\sum_{h \leq \eta_* N} \mathbb{E}[A_{N,h}^{\text{out}}] p_{N,h}(\delta_*) \leq CN^{2\gamma+1/2} \left[(1 + G_{B_N,1/2}(\rho_*))^{N/B_N} - 1 \right].$$

Using (15), the right side is bounded by a constant times

$$\frac{N^{2\gamma+3/2}}{B_N} \left(\frac{1 + \rho_*}{\sqrt{2}} \right)^{B_N}.$$

Finally, outward arithmetic gives

$$17 \log_2 \left(\frac{\sqrt{2}}{1 + \rho_*} \right) - (2\gamma + 3/2) > 0.0576243.$$

Thus the last expression tends to zero. $\square$

The hash-bound artifact named in Appendix A.1 contains the complete structured-route and RM2Sub transfer derivations, verifier sources, receipts, and arithmetic semantics. The theorem uses the probability space and inequalities stated in the main text and appendices; the artifact supplies the independently checkable arithmetic witnesses.